

Nakul Iyer

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EDUCATION

Purdue University

B.S. in Computer Engineering

West Lafayette, IN

Aug. 2025 – May 2028

- **Relevant Coursework:** Multivariate Calculus, Ordinary Differential Equations, Discrete Mathematics, Electrical Engineering Fundamentals I/II, Advanced C Programming, Python for Data Science

EXPERIENCE

Undergraduate Researcher | *Purdue VIP: Computer Vision & Generative Models*

Aug. 2026 – Present

- Developing a real time multimodal deep learning pipeline that combines 4 signal types, including facial, speech, lip synchronization, and acoustic features, to detect manipulated media in live video calls.
- Benchmarking audio and video deepfake models on 3 public datasets and benchmarks, including RTCFake, VCF, and FakeAVCeleb, using Slurm and NVIDIA GPUs on Purdue's Gilbreth cluster to test robustness against compression, packet loss, and noise suppression.

Design Lead | *Purdue Engineering Projects in Community Service*

Jan. 2026 — Present

- Lead a team of 8 students to develop a gamified early literacy application for the Boys & Girls Club of Northern Indiana Corridor, aligning structured learning modules and UI workflows with the UFLI phonics curriculum to support classroom and at-home literacy development.
- Engineer core application features using the Flutter + Firebase stack, including sound identification exercises, CVC word construction, guided tracing interfaces, diagnostic placement logic, and level-based reward progression; implemented scalable backend data models to track student progress and support adaptive learning pathways.

Software Developer | *Hack the Future*

Oct. 2025 — Present

- Built a centralized internal operations platform for Sleep Pod, a UK-based nonprofit distributing insulated sleeping pods across the UK and Europe, using the MERN stack (MongoDB, Express.js, React, Node.js) to replace fragmented WhatsApp, email, and spreadsheet workflows.
- Designed and implemented an analytics module using Apache ECharts with dual-party data verification, automated system-generated events for prison partners, and interactive visualizations for production amounts, material breakdowns, and geographic distribution.
- Built a real-time pending/history event pipeline using React, Node.js, Express.js, MongoDB, Stream Chat API, FullCalendar, and Apache ECharts, with role-based card views, dispute detection, admin override capabilities, and filtering by date, user, organization, and item type.

PROJECTS

Stochastic Portfolio Theory Risk Engine | *Python, NumPy, SciPy, pytest, Streamlit*

Aug. 2026 — Present

- Building a Monte Carlo VaR/ES engine generating 10,000+ path scenarios at 95%/99% confidence across four pluggable simulation backends (Cholesky-correlated Gaussian, historical bootstrap, and two rank-based models from Fernholz–Karatzas Stochastic Portfolio Theory), reporting confidence intervals on the VaR estimate itself.
- Encoded three closed-form theorem targets as failing tests before writing simulator code, with 200-case property tests per invariant; identified an off-diagonal covariance error in the leading open-source SPT implementation.

Gem Hunters Game | *Python, Javascript, JsPsych, Node.js*

June 2024 — Aug. 2024

- Built a web-based multidimensional reinforcement learning task in JavaScript using jsPsych where participants completed 216 trials each to learn reward rules that were changed between rounds.
- Engineered randomized experiment logic and data collection pipelines, implementing probabilistic reward schedules (80/20) and reaction time logging across 20,000+ trials, enabling precise measurement of learning performance.
- Deployed the experiment to 100+ participants via Prolific and analyzed response patterns, producing statistically usable data that was presented to 40+ professors, PhD students, and graduate researchers at the Mount Sinai.

TECHNICAL SKILLS

Languages: R, Python, Java, JavaScript, HTML/CSS, L^AT_EX, React, SQL, MATLAB

Libraries: Tidyverse, ggplot, jsPsych, NumPy, scikit-learn, Matplotlib, pandas, Node.js, Express.js

Tools: GitHub, Cursor, Visual Studio Code, MongoDB, Docker, Claude Code, CodeX, Jira